SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I L Charan Sai *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks
Concept Identification	20%	2	2	2
Concept Mapping Structure	25%	2	2	2
Linking & Relationships	20%	1	1	1
Application & Analysis	20%	1	1	1
Explanation & Clarity	15%	1	1	1

Total Marks 21 /30

Signature of the Faculty

CONCEPT MAPPING QUESTIONS

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3.Given the concept map:
DataLink→MACAddress→LocalDelivery
Network → IP Address → Global Delivery
Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:
ErrorControl
→DataLinkLayer→ErrorDetection
→ Transport Layer → Error Correction
Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:
Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗
Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

22/10/26.
Serial No: 22.

COI-AT2.

L. Charan Jai Reddy.
192526020
Computer Networks

ERROR CONTROL CONCEPT MAP.

Data link layer
Error Detection
- → uses CRC (cyclic Redundancy check) or check sum to detect error.
- → Identifies corrupted or lost data at the link level

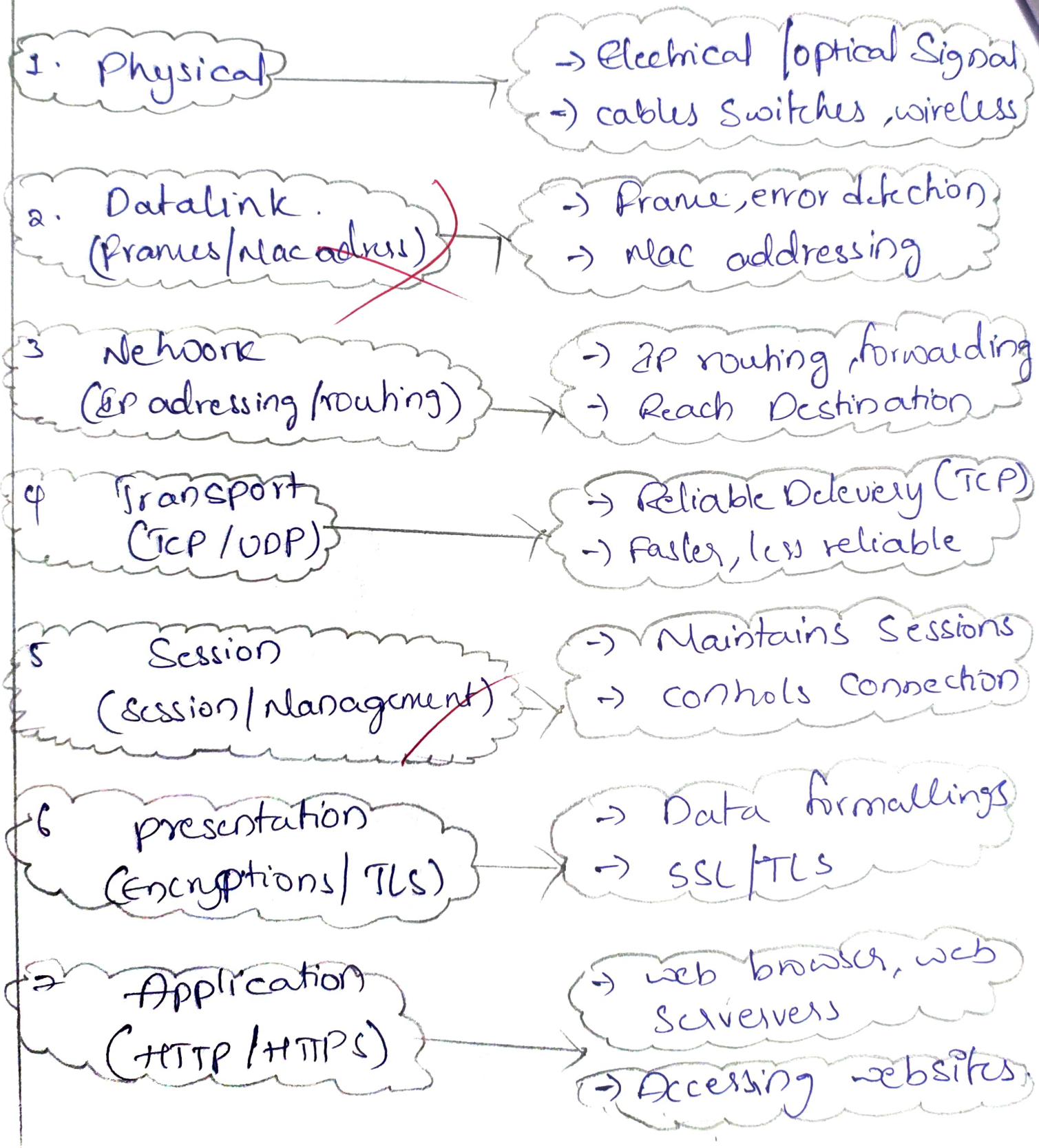
Transport Layer
Reliable delivery
- → uses Ack (Acknowledgment) and retransmission
- → Detect Missing or corrupted Data Segments.
- → Ensure End-to-End reliable communication

Reliable Data transmission

Analysis
- → The data link layer Detects error in frames using CRC or checksum.
- → The transport layer (eg TCP) Detect missing or corrupted Data and can retransmit

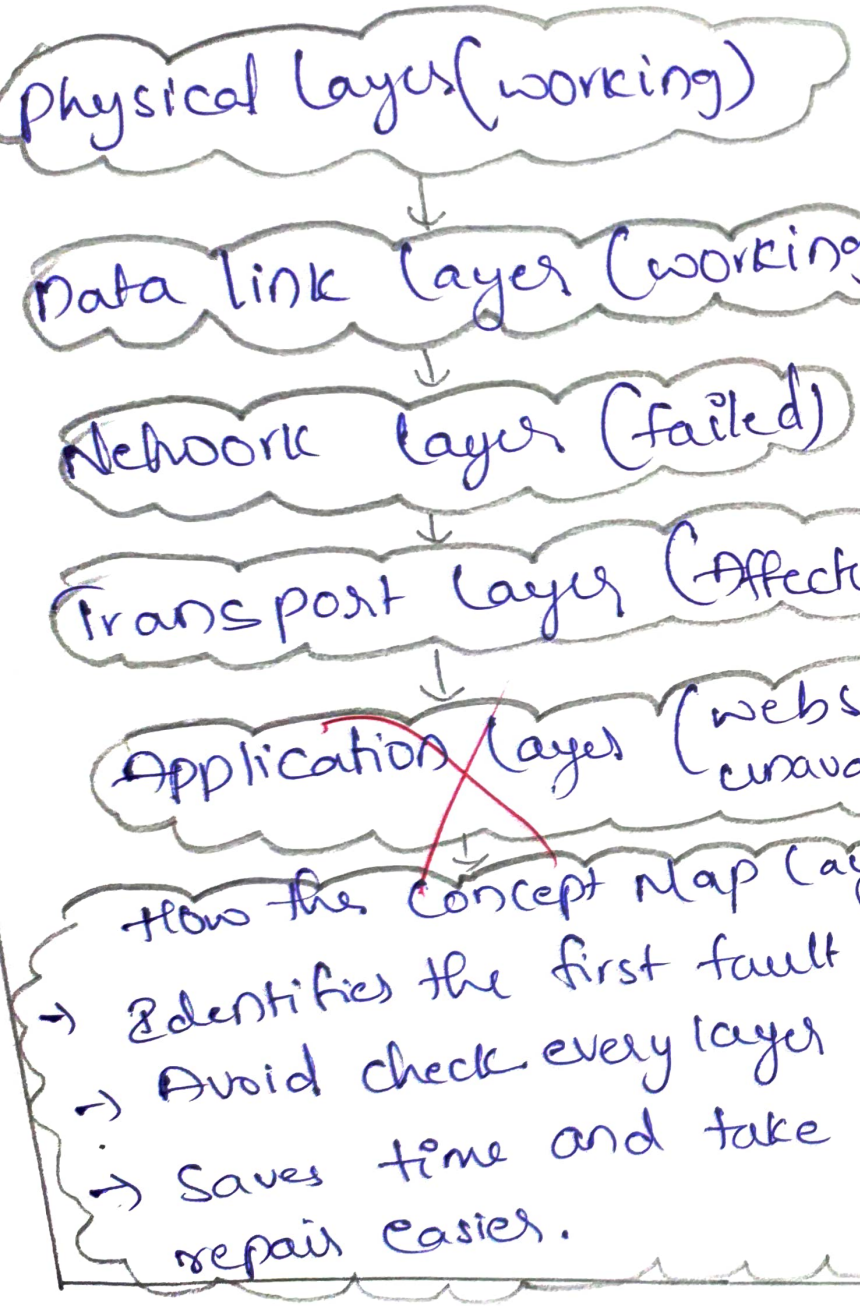
conclusion
Error Detection Identifies problem while transport level recovery helps deliver the correct data.

Web Browsing using OSI Layer concept Map.



Network Trouble Shooting

Concept Map.



Explanation.

1. Physical & Data layers are working (connections exists and local communication is fine
2. Network layer has failed
3. Transport and Application layer and affected because they depend on the Network layer.

How the Concept Map layer:

- → Identifies the first fault layer
- → Avoid check every layer unecessarily
- → Saves time and take Network repair easier.

Possible causes

- → Incorrect IP address
- → Wrong gateway
- → Route problem
- → Network configuration issue.